

RYTT Sovereign Semiotics:

1:1 Parallel Rosetta Stone Edition

DUAL-STREAM TRANSLATION: CANONICAL ENGLISH $\longleftrightarrow$ NATIVE GEOMETRIC
POLYTOPIC SEMIOTICS

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<https://github.com/Mega-Therion/Rytt-Sovereign-Semiotics>

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★ CANONICAL ENGLISH STREAM ★

★ NATIVE RYTT SEMIOTIC STREAM ★

1. Abstract & Theoretical Foundation

Modern computational linguistics and large neural architectures decouple statistical subword tokenization from symbolic geometry. Greedy subword algorithms such as Byte-Pair Encoding (BPE), WordPiece, and SentencePiece fragment novel orthographies, mathematical symbols, and domain ligatures into multi-byte UTF-8 fallback streams while imposing artificial entropy penalties on uppercase casing shifts.

We present **RYTT Sovereign Semiotics**, a complete geometric orthography, spatial coordinate algebra, and native integer chord compiler that unifies 26 fundamental geometric glyph primitives with lossless dual-plane casing mechanics. By assigning structural lowercase glyphs to the Ground Plane ($Z = 0.0$, Unicode PUA $\text{\uE000--\uE7FF}$) and emphatic uppercase glyphs to the Elevated Plane ($Z = 25.0$, Unicode PUA $\text{\uE800--\uEFFF}$), RYTT eliminates case-switching token penalties while preserving full semantic topology.

$\Sigma_M^\dagger \Gamma_{03} \Sigma_e \Sigma_r \Sigma_n \quad \Sigma_c \Gamma_{42} \Sigma_p \Sigma_u \Sigma_t \quad \Sigma_l \Sigma_i \Sigma_n \Sigma_g$
 $\Sigma_a \Sigma_n \Sigma_d \quad \Sigma_l \Sigma_a \Sigma_r \Sigma_g \Sigma_e \quad \Sigma_n \Sigma_e \Sigma_u \Sigma_r \Sigma_a \Sigma_l$
 $\Sigma_a \Sigma_r \Sigma_c \Sigma_h \quad \Sigma_d \Sigma_e \Sigma_c \Sigma_o \Sigma_u \Sigma_p \Sigma_i \Sigma_e \quad \Sigma_s \Sigma_t \Sigma_a \Sigma_t$
 $\Sigma_s \Sigma_u \Sigma_b \Sigma_w \Sigma_o \Sigma_r \Sigma_d \quad \Sigma_t \Sigma_o \Sigma_k \Sigma_e \Sigma_n$
 $\Sigma_f \Sigma_r \Sigma_o \Sigma_m \quad \Sigma_s \Sigma_y \Sigma_m \Sigma_b \Sigma_o \Sigma_l \Sigma_i \Sigma_c$
 $\Sigma_g \Sigma_e \Sigma_o \Sigma_m \Sigma_e \Sigma_t \Sigma_r \Sigma_y \cdot \quad \Sigma_G^\dagger \Sigma_r \Sigma_e \Sigma_e \Sigma_d \Sigma_y$
 $\Sigma_a \Sigma_l \Sigma_g \Sigma_o \quad \Sigma_B \Sigma_P \Sigma_E, \quad \Sigma_W \Sigma_o \Sigma_r \Sigma_d \Sigma_P,$
 $\Sigma_S \Sigma_e \Sigma_n \Sigma_t \Sigma_P \quad \Sigma_f \Sigma_r \Sigma_a \Sigma_g \Sigma_m \Sigma_e \Sigma_n \Sigma_t$
 $\Sigma_n \Sigma_o \Sigma_v \Sigma_e \Sigma_l \quad \Sigma_o \Sigma_r \Sigma_t \Sigma_h \Sigma_o \Sigma_g.$

$\Sigma_W^\dagger \Sigma_e \quad \Sigma_p \Sigma_r \Sigma_e \Sigma_s \Sigma_e \Sigma_n \Sigma_t$
 $\Sigma_R^\dagger \Sigma_Y^\dagger \Sigma_T^\dagger \Sigma_T^\dagger \quad \Sigma_S^\dagger \Sigma_o \Sigma_v \Sigma_e \Sigma_r \Sigma_e \Sigma_i \Sigma_g \Sigma_n$
 $\Sigma_S^\dagger \Sigma_e \Sigma_m \Sigma_i \Sigma_o \Sigma_t \Sigma_i \Sigma_c \Sigma_s, \quad \Sigma_g \Sigma_e \Sigma_o \Sigma_m \Sigma_e \Sigma_t \Sigma_r \Sigma_i \Sigma_c$
 $\Sigma_o \Sigma_r \Sigma_t \Sigma_h \Sigma_o \Sigma_g \Sigma_r \Sigma_a \Sigma_p \Sigma_h \Sigma_y, \quad \Sigma_s \Sigma_p \Sigma_a \Sigma_t \Sigma_i \Sigma_a \Sigma_l$
 $\Sigma_c \Sigma_o \Sigma_o \Sigma_r \Sigma_d \Sigma_i \Sigma_n \Sigma_a \Sigma_t \Sigma_e \quad \Sigma_a \Sigma_l \Sigma_g \Sigma_e \Sigma_b \Sigma_r \Sigma_a,$
 $\Sigma_c \Sigma_h \Sigma_o \Sigma_r \Sigma_d \quad \Sigma_e \Sigma_o \Sigma_m \Sigma_p \Sigma_i \Sigma_l \Sigma_e \Sigma_r \quad \Sigma_2 \Sigma_6$
 $\Sigma_g \Sigma_l \Sigma_y \Sigma_p \Sigma_h \quad \Sigma_p \Sigma_r \Sigma_i \Sigma_m \Sigma_i \Sigma_t \Sigma_i \Sigma_v \Sigma_e \Sigma_s.$
 $\Sigma_G \Sigma_r \Sigma_o \Sigma_u \Sigma_n \Sigma_d \quad (Z = 0.0, \text{\uE000}) \longleftrightarrow$
 $\Sigma_E^\dagger \Sigma_l \Sigma_e \Sigma_v \Sigma_a \Sigma_t \Sigma_e \Sigma_d \quad (Z = 25.0, \text{\uE800}).$
 $\Delta S_{\text{casing}} \equiv 0.$

The Bijective Involution Theorem ($\mathcal{D}(\mathcal{C}(S)) \equiv S$)

Let $\mathcal{A}$ be the complete alphabet and $\mathcal{S}^*$ be the free monoid of text strings. The RYTT encoding map $\mathcal{C} : \mathcal{S}^* \rightarrow \mathcal{V}^*$ and decoding map $\mathcal{D} : \mathcal{V}^* \rightarrow \mathcal{S}^*$ satisfy exact left-invertibility with zero topological distortion:

$$\forall S \in \mathcal{S}^*, \quad \mathcal{D}(\mathcal{C}(S)) = S$$

$\forall S \in \mathcal{S}^*, \quad \mathcal{D}(\mathcal{C}(S)) \equiv S.$
 $\Sigma_L^\uparrow \Sigma_e \Sigma_a \Sigma_n \ 4 \vdash \text{rytt_ground_primitive_left_inverse} :$
 $\Sigma_0 \text{ sorry.}$
 $\Sigma_L^\uparrow \Sigma_e \Sigma_a \Sigma_n \ 4 \vdash \text{rytt_elevated_compound_left_inverse} :$
 $\Sigma_0 \text{ sorry.}$

2. The 26-Primitive Coordinate Atlas

Glyph	ID	Plane	PUA	Class	Geometric Vector Construction
A	01	Ground / Elev	<code>\uE001 / \uE801</code>	$\mathcal{V}$	Diagonals $(15, 0) \rightarrow (50, 100) \rightarrow (85, 0)$; Bar $(30, 40) \rightarrow (70, 40)$
B	02	Ground / Elev	<code>\uE002 / \uE802</code>	$\mathcal{L} + \mathcal{C}$	Spine $(20, 0) \rightarrow (20, 100)$; Lobes $(20, 100) \curvearrowright (20, 50) \curvearrowright (20, 0)$
C	03	Ground / Elev	<code>\uE003 / \uE803</code>	$\mathcal{C}$	Smooth Open Manifold $(80, 85) \curvearrowright (20, 50) \curvearrowright (80, 15)$
D	04	Ground / Elev	<code>\uE004 / \uE804</code>	$\mathcal{L} + \mathcal{C}$	Spine $(20, 0) \rightarrow (20, 100)$; Single Convex Lobe $(20, 100) \curvearrowright (20, 0)$
E	05	Ground / Elev	<code>\uE005 / \uE805</code>	$\mathcal{L}$	Spine $(20, 0) \rightarrow (20, 100)$; Horizontals at $Y \in \{0, 50, 100\}$
M	13	Ground / Elev	<code>\uE00D / \uE80D</code>	$\mathcal{V}$	Spines $(15, 0) \rightarrow (15, 100), (85, 0) \rightarrow (85, 100)$; Apex $(15, 100) \rightarrow (50, 30) \rightarrow (85, 100)$
O	15	Ground / Elev	<code>\uE00F / \uE80F</code>	$\mathcal{O}$	Continuous Closed Jordan Curve $S^1 : (50, 10) \curvearrowright (15, 50) \curvearrowright (50, 90) \curvearrowright (85, 50)$
S	19	Ground / Elev	<code>\uE013 / \uE813</code>	$\mathcal{C}$	Serpentine Double-Inflexion Arc $(80, 85) \curvearrowright (50, 50) \curvearrowright (20, 15)$
T	20	Ground / Elev	<code>\uE014 / \uE814</code>	$\mathcal{L}$	Crosshead $(10, 100) \rightarrow (90, 100)$; Central Stem $(50, 100) \rightarrow (50, 0)$
V	22	Ground / Elev	<code>\uE016 / \uE816</code>	$\mathcal{V}$	Symmetrical Apex Chevron $(15, 100) \rightarrow (50, 0) \rightarrow (85, 100)$
X	24	Ground / Elev	<code>\uE018 / \uE818</code>	$\mathcal{X}$	Orthogonal Centroid Intersection $(15, 0) \rightarrow (85, 100), (15, 100) \rightarrow (85, 0)$

3. Dual-Plane Holonomic Integration

The RYTT token sequence integrates directly into a 4-tier holonomic polygon stacking kernel ($\Delta_3 \rightarrow 9\text{-gon} \rightarrow 7\text{-gon} \rightarrow 21\text{-gon}$) enabling real-time neural operator field evaluation and GPU cognitive raymarching:

$\Sigma_H^\uparrow \Sigma_o \Sigma_l \Sigma_o \Sigma_n \Sigma_o \Sigma_m \Sigma_i \Sigma_c \ \Sigma_K \Sigma_e \Sigma_r \Sigma_n \Sigma_e \Sigma_l : \quad \Delta_3 \rightarrow \Pi_9 \rightarrow \Pi_7 \rightarrow \Pi_{21}.$
 $\Sigma_R \Sigma_a \Sigma_y \Sigma_m \Sigma_a \Sigma_r \Sigma_c \Sigma_h : \quad 10,000+$
 $\Sigma_p \Sigma_a \Sigma_t \Sigma_h \Sigma_s / \Sigma_s \Sigma_c \Sigma_a \Sigma_n.$

$$E_n = \hbar \omega_0 \left(n + \frac{1}{2} \right) + \sum_{k=1}^4 \lambda_k \cos(k\theta)$$

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Empirical Benchmark Convergence

Across technical, philosophical, and codebase corpora, RYTT reduces token sequence length by 24.8%–31.2% compared to raw character streams with 100% mathematical invertibility.

$\Sigma_E^\uparrow \Sigma_m \Sigma_p \Sigma_i \Sigma_r \Sigma_i \Sigma_c \Sigma_a \Sigma_l \Sigma_R \Sigma_e \Sigma_s \Sigma_a \Sigma_l \Sigma_t \Sigma_s :$
 $\Sigma_C \Sigma_h \Sigma_a \Sigma_r \Sigma_s \rightarrow \Sigma_P \Sigma_U \Sigma_A : -16.0\%$ (Blake)
 $\Sigma_D \Sigma_i \Sigma_r \Sigma_a \Sigma_c \Sigma_Q \Sigma_M : -12.3\%$ (Physics)
 $\Sigma_E \Sigma_u \Sigma_l \Sigma_e \Sigma_r : -16.0\%$ (Math)
 $\Sigma_P \Sigma_y \Sigma_t \Sigma_h \Sigma_o \Sigma_n : -5.7\%$ (Code)
 $\Sigma_S^\uparrow \Sigma_t \Sigma_a \Sigma_t \Sigma_u \Sigma_s : \text{PASSED}$ (100% Lossless)